

40-Day Python Logic Mastery Plan

Phase 1: 23 - 31 July

PHASE 1: Build Foundation (23 - 31 July)

23 Jul: Variables, input(), print()

- Input name & age, print sentence
- Input 2 numbers, show sum
- Input name, show length

24 Jul: if, else, elif

- Check odd/even
- Age category: Child, Teen, Adult
- Divisible by 3 and 5?

25 Jul: for, while loops

- Print 1 to 10 using for & while
- Sum 1 to 50
- Table of user input

26 Jul: range(), nested loops, patterns

- 5x5 square of stars
- Triangle pattern
- Number triangle

27 Jul: Lists: basics

- List of fruits, update items
- Append new item, print length

28 Jul: Loops + Lists

- Even numbers from list

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- List sum, max (manual)

29 Jul: Retry 3 problems you failed

30 Jul: Journal - write struggles & wins

31 Jul: Mock Test (5 real logic Qs)

Phase 2: 1 - 10 August

PHASE 2: Boost Logic (1 - 10 Aug)

1 Aug: Strings

- Count vowels
- Reverse string
- Count 'a'

2 Aug: String Logic

- Palindrome check
- Capitalize first letter
- Replace 'a' with '@'

3 Aug: Lists Advanced

- Remove odds
- Manual sort
- Frequency count

4 Aug: Nested loops

- Star pyramid
- Number pyramid

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5 Aug: Edabit Easy Practice

6 Aug: Functions 1

- greet(), add(), square()

7 Aug: Functions 2

- Prime checker, factorial, vowel counter

8 Aug: Logic Quiz Day

9 Aug: Journal - logic review

10 Aug: Light practice + video

Phase 3: 11 - 20 August

PHASE 3: Core Skills (11 - 20 Aug)

11 Aug: Dictionary

- Store name-age pairs, update/delete

12 Aug: Set & Tuple

- Remove duplicates, access elements

13 Aug: File Handling

- Write, read, append to file

14 Aug: Try-Except

- Catch divide-by-zero, input errors

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15 Aug: Logic Day

- FizzBuzz, Armstrong, list primes

16 Aug: Retry Failed Questions

17 Aug: Function + Lists

- Return even numbers

18 Aug: Function + Strings

- Clean and count words

19 Aug: Mini-Test (5 Qs)

20 Aug: Journal: "What I Can Build"

Phase 4: 21 - 31 August

PHASE 4: Real Project Week (21 - 31 Aug)

21 Aug: Choose Project (BMI / ToDo / Expense Tracker)

22 Aug: Design app UI

23-26 Aug: Build project in parts

27 Aug: Polish with formatting, menus

28 Aug: Debugging Day

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29 Aug: Add comments + explain

30 Aug: Mock Quiz (5 questions)

31 Aug: Make your own certificate!



30 Days of Loop Logic & Condition Practice (Python)



Day-wise Challenges:

- Day 1 (Sunday) – Print a 5x5 square of stars.
- Day 2 – Print a right-angled triangle of stars (increasing pattern).
- Day 3 – Print an inverted right-angled triangle of stars.
- Day 4 – Print a triangle of numbers like: 1 12 123
- Day 5 – Each row starts from 1 up to the row number (like above).
- Day 6 – Print a pyramid pattern of stars.
- Day 7 – Print an inverted pyramid of stars.
- Day 8 – Print a diamond pattern of stars.
- Day 9 – Print a triangle of even numbers only.
- Day 10 – Print a triangle of alphabets (A, AB, ABC...).
- Day 11 – Floyd's Triangle: 1 2 3 4 5 6
- Day 12 – Print a checkerboard of stars and spaces.
- Day 13 – Print a right-aligned triangle of stars.
- Day 14 – Print a triangle with same number in a row (1, 22, 333...).
- Day 15 – Print a triangle with continuous numbers.

- **Day 16** – Print Pascal's Triangle (basic, with loops only).
- **Day 17** – Triangle of odd numbers only.
- **Day 18** – Half diamond pattern using stars: * ** *** ** *
- **Day 19** – Triangle with decreasing numbers in a row.
- **Day 20** – Triangle where each row starts with the row number.
- **Day 21** – Multiplication table of a number (use input).
- **Day 22** – Check if a number is prime using loop + condition.
- **Day 23** – Print all primes up to N.
- **Day 24** – Check if a number is a palindrome using loop.
- **Day 25** – FizzBuzz from 1 to 100.
- **Day 26** – Sum of digits of a number using loops.
- **Day 27** – Factorial of a number using loop.
- **Day 28** – Basic ATM simulation (loop until user exits).
- **Day 29** – Simulate 3 login attempts (loop + condition).
- **Day 30** – Print a number/star X-pattern (advanced).

